Akoizo Original Practice - Form 2

Division C | Experimental Design | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Which is a controlled variable rather than a control treatment?

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

A. A baseline reference level

B. An untreated comparison

C. Keeping light constant

D. A no-added-fertilizer group

Answer: _____

2. (2 points) A standard of comparison is…

Unit: Division C comparison standards, significant figures, and abstracts.
Worked-case data: Report area from measured lengths 2.0 cm and 3.00 cm, then report their sum.

A. A defined reference condition

B. Every controlled variable

C. Always a zero response

D. Always the largest treatment

Answer: _____

3. (2 points) Standard deviation has…

Unit: Quantitative data, graphs, and statistics.
Worked-case data: For response values 8, 10, 12, calculate mean, range, and sample SD.

A. The independent-variable units necessarily

B. The response’s units

C. Squared response units

D. No units always

Answer: _____

4. (2 points) A failed hypothesis means the experiment is invalid?

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

A. Only if data vary

B. Only if the graph is flat

C. No

D. Yes

Answer: _____

5. (2 points) A conclusion should answer…

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

A. The original measured question

B. Every possible mechanism

C. A preferred hypothesis regardless of data

D. A different untested outcome

Answer: _____

6. (4 points) Improve “release and time it.”

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

Answer: _____
_____
_____

7. (4 points) Name three dependent reporting tasks.

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

Answer: _____
_____
_____

8. (4 points) Give a variance mechanism for timing.

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

Answer: _____
_____
_____

9. (4 points) Why use several levels?

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

Answer: _____
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_____

10. (4 points) Write a compact abstract skeleton.

Unit: Division C comparison standards, significant figures, and abstracts.
Worked-case data: Report area from measured lengths 2.0 cm and 3.00 cm, then report their sum.

Answer: _____
_____
_____

11. (4 points) Why can precise measurements be inaccurate?

Unit: Quantitative data, graphs, and statistics.
Worked-case data: For response values 8, 10, 12, calculate mean, range, and sample SD.

Answer: _____
_____
_____

12. (6 points) Solve this case and explain your reasoning: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

Answer: _____
_____
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13. (6 points) Solve this case and explain your reasoning: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

Answer: _____
_____
_____

14. (2 points) Define “Qualitative data” in the context of Experimental Design.

Answer: _____
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15. (2 points) Define “Bottleneck” in the context of Experimental Design.

Answer: _____
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16. (2 points) Define “Evidence” in the context of Experimental Design.

Answer: _____
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17. (2 points) Define “Replication” in the context of Experimental Design.

Answer: _____
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18. (2 points) Define “Significant figures” in the context of Experimental Design.

Answer: _____
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